Total No. of Questions : 08]

SEAT No. :

P645

[Total No. of Pages : 2

[5869]-272
S.E. (Computer/AIDS) (Semester III)
FUNDAMENTALS OF DATA STRUCTURES
(2019 Pattern)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8 .***
2) ***Figures to the right indicate full marks***
3) ***Neat diagrams must be drawn whenever necessary.***
4) ***Make suitable assumption whenever necessary.***

Q1) a) Write an algorithm of Bubble sort & sort the following numbers using Bubble sort & show the contents of an array after Every pass. **[9]**

81, 5, 27, –6, 61, 93, 4, 8, 104, 15

b) Explain the radix sort . Sort the following numbers in ascending order. **[9]**

14, 1, 66, 74, 22, 36, 41, 59, 64, 54

Obtain the time & space complexity of your algorithm.

OR

Q2) a) Explain internal & external sorting by taking suitable example of each type. **[9]**

b) Write a short note on Fibonacci search with suitable example. **[9]**

Q3) a) Write pseudo C++ code for addition of two polynomials using singly linked list. **[9]**

b) What is dynamic data structure? Explain the circular linked list with its basic operations. **[9]**

OR

Q4) a) Write a pseudo code for the addition of a node after the position ‘P’ in singly linked list. **[9]**

b) Explain the doubly linked list with it’s basic operations; list the advantages of doubly linked list over singly linked list. **[9]**

P.T.O.

Q5) a) Write a pseudo code for basic operations of stock. **[8]**

b) What are the variants of recursion, explain with example. **[9]**

OR

Q6) a) Explain the linked implementation of stock with suitable example. **[8]**

b) Write pseudo code for infix to postfix expression; Explain the need of conversion of expression. **[9]**

Q7) a) Define the following terms with example **[8]**

i) Linear queue

ii) Circular queue

iii) priority queue

b) Write pseudo C++ code to implement linked queue. **[9]**

OR

Q8) a) Explain array implementation of priority queue with all basic operation. **[8]**

b) Write a pseudo C++ code to implement Circular queue using array. **[9]**